

HW4

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Skolem Normal Form and Skolemization

Definition 0.1 (Skolem Normal Form). *A formula is in Skolem Normal Form if it is a sentence in prenex normal form with no existential quantifiers.*

Definition 0.2 (Skolemization). *Let φ be a FOL formula. Its Skolem Normal Form $\text{sk}(\varphi)$ is obtained as follows:*

1. *Convert φ to an equivalent prenex normal form ψ over the same signature.*
 2. *Existentially quantify all free variables to obtain a sentence ψ_0 .*
 3. *Replace each existential variable x by a Skolem term:*
 - *If $\text{dep}(x) = \emptyset$, replace x by a fresh constant c_x .*
 - *Otherwise, replace x by a fresh function $f_x(\bar{y})$ where $\bar{y} = \text{dep}(x)$.*
- Remove all existential quantifiers.*

General Form

$$\psi_0 \equiv \exists x_{1,1} \cdots \exists x_{1,k_1} \forall y_{1,1} \cdots \forall y_{1,l_1} \cdots \exists x_{m,1} \cdots \exists x_{m,k_m} \forall y_{m,1} \cdots \forall y_{m,l_m} \psi_b$$

$$\text{sk}(\varphi) = \forall y_{1,1} \cdots \forall y_{1,l_1} \cdots \forall y_{m,1} \cdots \forall y_{m,l_m} \psi_b[x_{i,j} \mapsto \text{sTrm}(x_{i,j})]$$

Properties

- $\text{sk}(\varphi)$ is a sentence
- $\text{sk}(\varphi)$ has no existential quantifiers
- $\text{sk}(\varphi)$ is over an extended signature

Question 1

Show that for any FOL formula φ ,

$$\varphi \text{ is satisfiable} \iff \text{sk}(\varphi) \text{ is satisfiable.}$$

Proof. Consider skolemization as a recursive algorithm described as follows:

Algorithm 1: Skolemization on Prenex Formula

Input: Signature $\Sigma = (C, F, R)$;
Formula $\phi = (\pi, \psi)$ where π is a quantifier prefix and ψ is quantifier-free;
List u of universally quantified variables in scope (initially empty).
Output: Skolemized formula (π', ψ') equisatisfiable with ϕ .
if π *is empty* **then**
 return $(\emptyset, \psi)$
 $(q, x) \leftarrow \text{head}(\pi)$;
 $\pi_{\text{rest}} \leftarrow \text{tail}(\pi)$
if $q = \exists$ **then**
 if u *is empty* **then**
 introduce fresh constant c_x ;
 $\Sigma.C \leftarrow \Sigma.C \uplus \{c_x\}$;
 return $\text{Skol}(\Sigma, (\pi_{\text{rest}}, \psi[x \mapsto c_x]), u)$
 else
 introduce fresh function f_x of arity $|u|$;
 $\Sigma.F \leftarrow \Sigma.F \uplus \{f_x\}$;
 let $t = f_x(u_1, \dots, u_k)$ where $u = [u_1, \dots, u_k]$;
 return $\text{Skol}(\Sigma, (\pi_{\text{rest}}, \psi[x \mapsto t]), u)$
else
 // $q = \forall$
 $u' \leftarrow u \cdot x$ // **append** x **to** u
 $(\pi', \psi') \leftarrow \text{Skol}(\Sigma, (\pi_{\text{rest}}, \psi), u')$;
 return $((\forall x) \cdot \pi', \psi')$

We claim that the invariant is the satisfiability of input and output is the same. Let $\text{Skol}(\Sigma, (\pi, \psi), u)$ be the recursive algorithm defined. We prove by induction on the length of the quantifier prefix $n = |\pi|$ that the algorithm preserves equisatisfiability.

Inductive hypotshis

For any prefix π of length n , any signature Σ , and any list of universal variables $u = [u_1, \dots, u_k]$, a formula $\phi = \forall u(\pi\psi)$ is satisfiable in a Σ -structure $\mathfrak{A}$ if and only if there exists a larger $\mathfrak{A}'$ than $\mathfrak{A}$ to Σ' (containing new symbols) such that $\mathfrak{A}'$ satisfies the output of $\text{Skol}(\Sigma, (\pi, \psi), u)$.

$$\mathfrak{A} \models \phi = \forall u(\pi\psi) \iff \mathfrak{A}' \models \text{Skol}(\Sigma, (\pi, \psi), u)$$

Case 1: Base Case ($n = 0$)

If π is empty, the algorithm returns $(\emptyset, \psi)$, trivially equisatisfiable

Case 2: Universal Case ($q = \forall$)

Let $\pi = \forall x \cdot \pi_{\text{rest}}$. The algorithm sets $u' = u \cdot x$ and recurses

We will show that:

$$\mathfrak{A} \models \forall u \forall x (\pi_{\text{rest}} \psi) \iff \mathfrak{A}' \models \forall u \forall x (\text{Skol}(\pi_{\text{rest}}, \psi, u'))$$

- $(\Rightarrow)$: Suppose $\mathfrak{A} \models \forall u \forall x (\pi_{\text{rest}} \psi) \implies \mathfrak{A} \models \forall (u \cup \{x\}) (\pi_{\text{rest}} \psi)$. There exists an expansion $\mathfrak{A}'$ such that $\mathfrak{A}' \models \forall (u \cup \{x\}) \text{Skol}(\pi_{\text{rest}}, \psi, u')$. rearrange the quantifiers gives $\mathfrak{A}' \models \forall u \forall x \text{Skol}(\dots)$.
- $(\Leftarrow)$: Suppose $\mathfrak{A}' \models \text{Skol}(\pi_{\text{rest}}, \psi, u')$. Since $\mathfrak{A}'$ is an expansion of $\mathfrak{A}$, and the universal quantifier $\forall x$ does not introduce new symbols in this step, same logic

Case 3: Existential Case ($q = \exists$)

□

Theorem 1 (Restricted Downward Löwenheim-Skolem)

Let Σ be a countable signature and Γ a set of universally quantified FOL sentences without equality. If Γ is satisfiable, then there exists a countable structure $\mathfrak{A}$ such that

$$\mathfrak{A} \models \Gamma.$$

Theorem 2 (Full Downward Löwenheim-Skolem)

Let Σ be a countable signature, V a countable set of variables, and Γ a set of FOL formulae. If Γ is satisfiable, then there exists a countable structure $\mathfrak{A}$ such that

$$\mathfrak{A} \models \Gamma.$$

Question 2

Prove Theorem 2.

Proof. Let Γ be a set of FOL formulae over a countable signature Σ .

1. For each $\phi \in \Gamma$ with free variables $x_1, \dots, x_n$, let $\phi' = \exists x_1 \dots \exists x_n \phi$. Γ is sat $\iff \Gamma' = \{\phi' \mid \phi \in \Gamma\}$ is satisfiable
2. Convert Γ' to $sk(\Gamma)$ by skolemization. $sk(\Gamma)$ is a set of universal sentences over an extended countable signature Σ' . By Question 1, $sk(\Gamma)$ is sat $\iff \Gamma'$ is sat
3. Since $sk(\Gamma)$ consists of universal sentences, by Theorem 1, there exists a countable structure $\mathfrak{A} = (U, I)$ such that $\mathfrak{A} \models sk(\Gamma)$
4. If Σ contains equality, treat $=$ as a binary relation EQ . By the congruence proof from hw4, define $I(EQ)$ to make a congruence relation on $\mathfrak{A}$.
5. construct a new quotient structure $\mathfrak{A}_{EQ}$ with universe $U_{EQ} = \{[u]_{EQ} \mid u \in U\}$ where equal elements collapse to the same equivalent class
6. $|U|$ is countable $\implies |U/EQ| \leq |U|$ is countable. By congruence, $\mathfrak{A}/EQ \models sk(\Gamma) \implies \mathfrak{A}/EQ \models \Gamma$.

□

Theorem 3 (Compactness)

Let Σ be a countable signature and Γ a set of FOL formulae. Then:

$$\Gamma \text{ is satisfiable} \iff \Gamma \text{ is finitely satisfiable.}$$

Question 3

Prove Theorem 3.

Definability

Definition 0.3. Let Σ be a signature and C a class of Σ -structures. C is definable in FOL if there exists a sentence φ such that

$$\llbracket \varphi \rrbracket = \{\mathfrak{A} \mid \mathfrak{A} \models \varphi\} = C.$$

Question 4 (Reachability)

Prove that reachability is not definable in FOL, i.e., there is no sentence φ_{reach} such that

$$\{G(\mathfrak{A}) \mid \mathfrak{A} \models \varphi_{\text{reach}}\} = G_{\text{reach}}.$$

Proof of Question 4. To prove that reachability is not definable in First-Order Logic (FOL), we proceed by contradiction using the Compactness Theorem

1. Let Σ_{graphs} be the signature $(\{s, t\}, \emptyset, \{E\})$ For each $k \in \mathbb{N}$, define a FOL formula $\psi_k(s, t)$ asserting there exists a path of length at most k from s to t :

$$\psi_k(s, t) \equiv \exists v_0 \dots \exists v_k \left((v_0 = s) \wedge (v_k = t) \wedge \bigwedge_{0 \leq i < k} E(v_i, v_{i+1}) \right)$$

2. Suppose there exists a sentence ϕ_{reach} such that $\mathfrak{A} \models \phi_{\text{reach}}$ iff t is reachable from s in $G(\mathfrak{A})$ Consider the set of sentences:

$$\Gamma = \{\phi_{\text{reach}}\} \cup \{\neg\psi_k(s, t) \mid k \in \mathbb{N}\}$$

3. Let $\Gamma_0 \subseteq \Gamma$ be any finite subset. There exists a maximum k_{max} such that $\neg\psi_{k_{\text{max}}}$ is in Γ_0 . Construct a graph G where t is reachable from s via a path of length $k_{\text{max}} + 1$. In this structure, ϕ_{reach} is true, and all $\neg\psi_k$ for $k \leq k_{\text{max}}$ are true. Thus, Γ_0 is finitely satisfiable.
4. By the Compactness Theorem, if every finite subset of Γ is satisfiable, then Γ is satisfiable. However, a model $\mathfrak{A} \models \Gamma$ would imply:

- t is reachable from s (since $\mathfrak{A} \models \phi_{\text{reach}}$)
- There is no finite path from s to t (since $\mathfrak{A} \models \neg\psi_k$ for all k)

contradiction, as reachability requires a finite sequence of vertices. Therefore, no such sentence ϕ_{reach} exists. □

Question 5 (k -Colorability)

Define a FOL sentence $\varphi_{k\text{-colorable}}$ such that

$$\{G(\mathfrak{A}) \mid \mathfrak{A} \models \varphi_{k\text{-colorable}}\} = G_{k\text{-colorable}}.$$

Proof. To define k -colorability over the signature $\Sigma_{\text{colored-graphs}} = (\emptyset, \emptyset, \{E, R_1, \dots, R_k\})$, we construct a sentence $\phi_{k\text{-colorable}}$ that enforces the following constraints:

1. Total Assignment:

$$\forall x \bigvee_{i=1}^k R_i(x)$$

2. Mutual Exclusivity:

$$\forall x \bigwedge_{1 \leq i < j \leq k} \neg(R_i(x) \wedge R_j(x))$$

3. Edge Consistency:

$$\forall x \forall y E(x, y) \implies \bigwedge_{i=1}^k \neg(R_i(x) \wedge R_i(y))$$

$$\phi_{k\text{-colorable}} \equiv \forall x \left(\bigvee_{i=1}^k R_i(x) \right) \wedge \forall x \left(\bigwedge_{i < j} \neg(R_i(x) \wedge R_j(x)) \right) \wedge \forall x \forall y \left(E(x, y) \implies \bigwedge_{i=1}^k \neg(R_i(x) \wedge R_i(y)) \right)$$

□

Definition 0.4. Let $\Sigma = (C, F, R)$ and $\mathfrak{A}_1, \mathfrak{A}_2$ be Σ -structures. A function $h : U_1 \rightarrow U_2$ is a homomorphism if:

1. $h(I_1(c)) = I_2(c)$ for all $c \in C$
2. $h(I_1(f)(\bar{u})) = I_2(f)(h(\bar{u}))$ for all $f \in F$
3. $(\bar{u} \in I_1(R)) \iff (h(\bar{u}) \in I_2(R))$ for all $R \in R$

Question 6

1. Prove $h(n) = n \bmod 2$ is a homomorphism from $\mathfrak{A}_{\mathbb{N}}$ to $\mathfrak{A}_{\mathbb{B}}$.
2. Find φ such that $\mathfrak{A}_{\mathbb{N}} \models \varphi$ but $\mathfrak{A}_{\mathbb{B}} \not\models \varphi$.

6.1

Proof. To prove $h : \mathbb{N} \rightarrow \mathbb{B}$ is a homomorphism from $\mathfrak{A}_{\mathbb{N}}$ to $\mathfrak{A}_{\mathbb{B}}$:

1. Constants:

- $h(I_{\mathbb{N}}(\mathbb{O})) = h(0) = 0 = I_{\mathbb{B}}(\mathbb{O})$.
- $h(I_{\mathbb{N}}(\mathbb{1})) = h(1) = 1 = I_{\mathbb{B}}(\mathbb{1})$.

2. Functions (+): We must show $h(m+n) = I_{\mathbb{B}}(+)(h(m), h(n))$ for all $m, n \in \mathbb{N}$.

- Case 1: m, n both even. $m+n$ is even. $h(m+n) = 0$. $I_{\mathbb{B}}(+)(0, 0) = 0$.
- Case 2: m, n both odd. $m+n$ is even. $h(m+n) = 0$. $I_{\mathbb{B}}(+)(1, 1) = 0$.
- Case 3: One even, one odd. $m+n$ is odd. $h(m+n) = 1$. $I_{\mathbb{B}}(+)(0, 1) = 1$.

□

6.2

Proof.

$$\phi \equiv \neg(1 + 1 = 0)$$

- $1 + 1 = 2 \neq 0 \implies \mathfrak{A}_{\mathbb{N}} \models \phi$.
- $1 + 1 = 0 \implies \mathfrak{A}_{\mathbb{B}} \models \neg\phi$.

□

Definition 0.5. A homomorphism $h : U_1 \rightarrow U_2$ is an isomorphism if it is bijective.

$$\mathfrak{A}_1 \cong \mathfrak{A}_2 \iff \exists h \text{ isomorphism}$$

Question 7

Prove that if $\mathfrak{A}_1 \cong \mathfrak{A}_2$, then $\mathfrak{A}_2 \cong \mathfrak{A}_1$.

Proof. Let $h : U_1 \rightarrow U_2$ be an isomorphism from $\mathfrak{A}_1$ to $\mathfrak{A}_2$. By definition, h is a bijective homomorphism. Since h is a bijection, there exists an inverse function $h^{-1} : U_2 \rightarrow U_1$ which is also a bijection. We show h^{-1} is a homomorphism:

1. $\forall c \in C$, we know $h(I_1(c)) = I_2(c)$. Applying h^{-1} to both sides, we get $I_1(c) = h^{-1}(I_2(c))$.
2. $\forall f \in F$ of arity k and $v_1, \dots, v_k \in U_2$, let $u_i = h^{-1}(v_i)$.

$$h(I_1(f)(u_1, \dots, u_k)) = I_2(f)(h(u_1), \dots, h(u_k)) = I_2(f)(v_1, \dots, v_k)$$

Applying h^{-1} on both sides:

$$I_1(f)(h^{-1}(v_1), \dots, h^{-1}(v_k)) = h^{-1}(I_2(f)(v_1, \dots, v_k))$$

3. $\forall R \in \mathcal{R}$ of arity k :

$$\begin{aligned} (v_1, \dots, v_k) \in I_2(R) &\iff (h(h^{-1}(v_1)), \dots, h(h^{-1}(v_k))) \in I_2(R) \\ &\iff (h^{-1}(v_1), \dots, h^{-1}(v_k)) \in I_1(R) \end{aligned}$$

Since h^{-1} is a bijective homomorphism, it is an isomorphism from $\mathfrak{A}_2$ to $\mathfrak{A}_1$. □

Question 8

1. Give examples of isomorphic structures.
2. Give examples of non-isomorphic structures with same finite cardinality.

8.1

Proof. Natural number to even natural number, shown in class □

8.2

Proof. $\mathfrak{A}_1$ and $\mathfrak{A}_2$ from question 6, just change $U_{\mathfrak{A}_1}$ from $\mathbb{N}$ to $\{0, 1\}$, redefine $+$ such that $1 + 1 = 1$ □

Theorem 4 (Isomorphism Theorem)

Let $h : \mathfrak{A}_1 \rightarrow \mathfrak{A}_2$ be an isomorphism. Then for every formula φ and assignment α ,

$$\mathfrak{A}_1, \alpha \models \varphi \iff \mathfrak{A}_2, (h \circ \alpha) \models \varphi.$$